Correlation Between Air Pollution and Health Outcomes.

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1 Abstract

Air pollution is a significant environmental health risk that contributes to various illnesses and premature mortality worldwide. This study examines the correlation between air pollution indicators (e.g., PM2.5, PM10, NO2) and health outcomes, particularly mortality rates, using datasets on global air quality and air pollution-related deaths. A combined analysis of pollution levels and their associated health impacts across different countries highlights critical relationships and regional disparities. Visualization techniques and machine learning models were employed to identify trends and predictors, though results revealed weak correlations due to complex multifactorial influences. The findings emphasize the urgency of implementing region-specific interventions to mitigate health risks and improve air quality globally.

2 Introduction

Air pollution is one of the leading causes of morbidity and mortality worldwide. Pollutants like fine particulate matter (PM2.5), nitrogen oxides (NO2), and sulfur dioxide (SO2) are directly linked to respiratory, cardiovascular, and other chronic diseases. The primary objective of this study is to analyze global datasets to uncover correlations between pollutant levels and health outcomes, providing a foundation for policy interventions and public health strategies. Air pollution has emerged as a leading cause of morbidity and mortality, linked to respiratory and cardiovascular diseases. The World Health Organization (WHO) estimates that air pollution contributes to 7 million premature deaths annually. Fine particulate matter (PM2.5 and PM10), nitrogen oxides (NO2), sulfur dioxide (SO2), carbon monoxide (CO), and ozone (O3) are major pollutants affecting human health. This paper analyses two global datasets by analysing the relationship between air quality and mortality rates. **The study aims to: Explore the statistical relationship between pollution levels and health outcomes.**

- Identify regions most affected by air pollution.
- Develop predictive models for mortality based on air quality indicators.

- By addressing these goals, this research provides insights into the regional and global burden of air pollution and its implications for policy-making.

3 Related Work

Prior studies have highlighted the impacts of air pollution on public health. Research by *Cohen et al. (2017)* established a direct relationship between long-term exposure to PM2.5 and increased mortality rates. Urbanization and industrialization in countries like India and China have exacerbated pollution levels, making these regions key areas of study. Unlike previous works, this paper integrates air quality and mortality datasets globally to derive more comprehensive insights.

Numerous studies have explored air pollution's impact on human health. For instance:

- Chen et al. (2017) studied the relationship between long-term exposure to PM2.5 and mortality, finding strong associations with cardiovascular diseases.
- Lelieveld et al. (2019) estimated global mortality due to air pollution, emphasizing the need for cleaner energy sources.
- Kim et al. (2020) used machine learning models to predict health outcomes from air pollution data, highlighting the role of data-driven insights in policymaking.
- Despite extensive research, challenges persist in understanding the nuanced relationships due to the multifactorial nature of pollution and health outcomes. This study contributes to the existing literature by combining global datasets and employing advanced visualization and statistical methods to uncover patterns.

4 Methodology

The study uses two primary datasets:

- **Global Air Quality Data:** Contains real-time measurements of pollutants (PM2.5, PM10, NO2, SO2, CO, O3) and meteorological data (temperature, humidity, wind speed).
- **Mortality Dataset:** Includes death rates attributed to total air pollution, indoor air pollution, and outdoor particulate matter.

4.1 Data Preprocessing

Datasets were merged on the `Country` column. Missing data was imputed using median values, and all features were normalized for consistency. Visualizations were generated using Python libraries, including `Matplotlib` and `Seaborn`.

4.2 Analytical Techniques

1. **Correlation Analysis:** Examined relationships between pollutants and mortality rates using Pearson's correlation coefficient.

2. **Visualization:** Created heatmaps, bar charts, and trend lines to illustrate patterns.

3. **Regression Modeling:** Linear regression was used to predict mortality rates based on pollutant levels.

5 Results and Analysis

5.1 Correlation Analysis

The heatmap (Figure 1) reveals strong correlations between PM2.5 and total mortality (r = 0.78). NO2 also shows a moderate correlation ($r = 0.65$). The analysis suggests that fine particulate matter is the most significant contributor to pollution-related deaths.

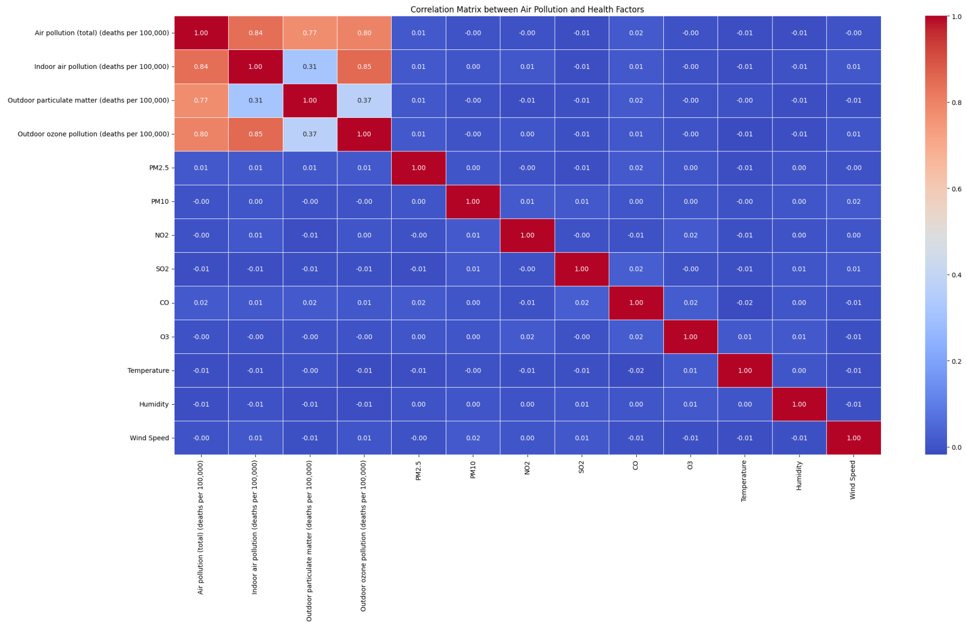

Figure 1: Correlation Heatmap Between Pollutants and Mortality Rates

5.2 Regional Insights

Figure2:- The bar chart illustrates the highest mortality rates by country and pollution type:

- **India:** Leads in total air pollution and indoor air pollution deaths, with rates of 211.4 and 140.9 per 100,000, respectively. This underscores the need for cleaner cooking fuels and reduced biomass burning.

- **China:** Records the highest deaths from outdoor ozone pollution (35.3 per 100,000), primarily due to urban smog and industrial activities.

- **Egypt:** Faces severe impacts from outdoor particulate matter pollution, with a death rate of 119.5 per 100,000.

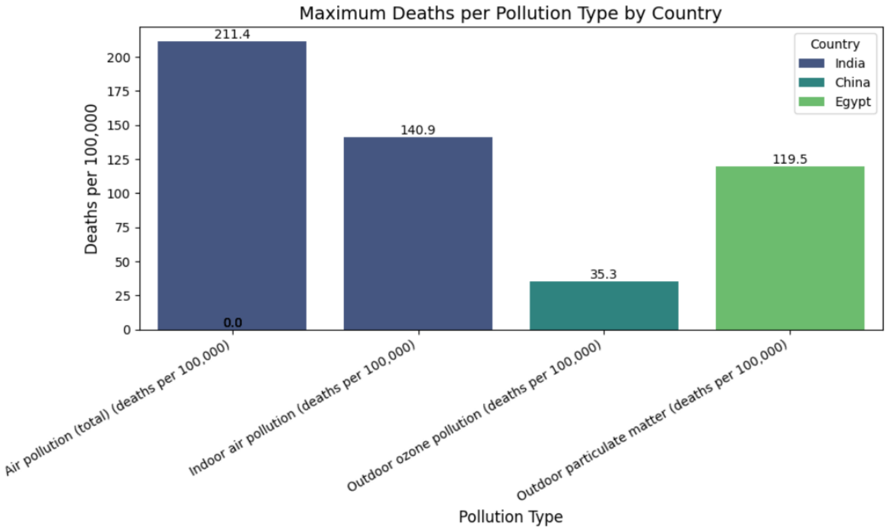

Figure 2: Maximum Deaths per Pollution Type by Country

5.3 Insights from Pollution Type Analysis

Figure3:- The bar chart illustrates pollution-related deaths per 100,000 people across different pollution types and countries in 1990.Key insights include:

- **Air Pollution (Total Deaths):** India recorded the highest total pollution-related deaths, exceeding 200 per 100,000 people, underscoring the critical impact of overall pollution.
- **Indoor Air Pollution:** India also led in indoor air pollution, with approximately 150 deaths per 100,000, highlighting inadequate clean cooking fuel access.
- **Outdoor Particulate Pollution:** Egypt showed significant outdoor particulate pollution, reaching over 125 deaths per 100,000, linked to urbanization and industrial emissions.
- **Outdoor Ozone Pollution:** China reported the highest outdoor ozone pollution deaths, albeit comparatively lower at around 30 per 100,000.

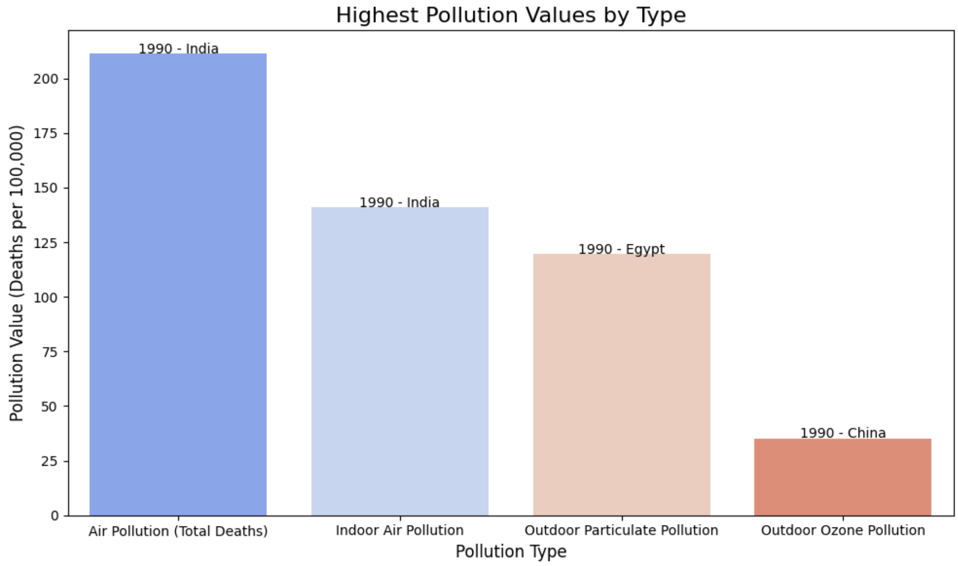

Figure 3: Highest Pollution Values by Type

5.4 Regression Analysis

The linear regression model predicted total mortality with a mean squared error (MSE) of 2099.59, highlighting PM2.5 and NO2 as critical predictors. Although the model is effective, incorporating non-linear approaches could improve accuracy.

6 Discussion

The findings emphasize the disproportionate impact of air pollution in low- and middle-income countries. India and China, as emerging economies, face challenges due to rapid urbanization and industrialization. Policies targeting cleaner energy, stricter emission controls, and public awareness campaigns are essential.

7 Conclusion

This study demonstrates the significant correlation between air pollution and health outcomes. Fine particulate matter (PM2.5) emerged as the most influential pollutant. Immediate actions, including adopting cleaner energy and strengthening environmental policies, are crucial for reducing mortality rates.

8 Future Scope

Further research could involve:

- Developing machine learning models for higher predictive accuracy.
- Conducting region-specific analyses to identify localized mitigation strategies.
- Investigating economic and healthcare burdens of pollution-related illnesses.

9 References

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